

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date:	03 October 2026
Team ID:	SWTID-2026-3257
Project Name:	Implement Client Script & UI Policy (Incident Table)
Maximum Marks:	4 Marks

1. Technical Architecture Overview

The Incident Management Client Script & UI Policy application follows a modern multi-tiered ITSM architecture. Client-side interactions occur within the ServiceNow / Web Browser Workspace, where Client Scripts and UI Policies execute natively in JavaScript to control form field visibility, mandatory requirements, and read-only states dynamically before sending data over secure HTTPS/REST APIs to the cloud platform application engine and relational database layer.

System Interaction Flow:

1. Service Desk Analyst accesses Incident Workspace via Web Browser (Client Layer).
2. On-Load & On-Change Events trigger Client Scripts & UI Policy Rules in the local browser DOM engine.
3. Real-time field manipulation (e.g., hiding Resolution Notes until State=Resolved) is executed client-side with zero server latency.
4. Validated incident data is transmitted over TLS 1.3 encrypted REST APIs to the Application Engine (ServiceNow Platform / Node Backend).
5. Record state and audit history are committed to the Relational Cloud Database (MariaDB / MySQL enterprise cluster).

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface (UI / Workspace)	Next-Gen Service Operations Workspace / Incident Form UI where support analysts interact with incident records.	ServiceNow Agent Workspace, HTML5, CSS3, JavaScript (ES6+), GlideForm (g_form) API
2	Application Logic - Client Side (UI Policies)	Declarative logic controlling dynamic field visibility, mandatory attributes, and read-only conditions based on state changes.	ServiceNow UI Policies, Client Script Engine, JavaScript DOM manipulation
3	Application Logic - Client Side	Real-time client-side event listeners (onChange,	JavaScript, GlideForm (g_form),

	(Scripts)	onLoad, onSubmit) for dynamic calculations and interactive prompts.	GlideUser (g_user), AJAX (GlideAjax)
4	Application Logic - Server Side	Platform application engine handling business rules, workflow orchestration, and server-side data validation.	ServiceNow Glide Engine, Rhino JavaScript Engine / Node.js runtime
5	Database Layer	Relational database storing Incident record history, user roles, assignment groups, and audit trails.	MariaDB / MySQL enterprise cloud instance (ServiceNow Cloud Data Tier)
6	Cloud Platform / Storage	Cloud platform infrastructure hosting the ITSM application instance and document attachments.	ServiceNow Multi-Instance Enterprise Cloud / AWS Cloud Storage
7	External API - Incident Integration	RESTful web services for integrating incident data with third-party monitoring systems or notification systems.	REST API, JSON, OAuth 2.0 Webhooks
8	Machine Learning / Analytics	Predictive intelligence and auto-categorization model for incident ticket routing.	Predictive Intelligence ML Engine / Python scikit-learn (Optional)
9	Infrastructure & Deployment	Cloud-hosted multi-tenant enterprise instance running on high-availability cluster architecture.	Cloud Platform, Docker / Kubernetes container runtime for custom microservices

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source & Standard Frameworks	Standard web technologies and JavaScript libraries used for client-side rendering and scripting.	JavaScript (ES6+), HTML5, CSS3, REST API standards, JSON data interchange
2	Security Implementations	Role-based access control (RBAC), secure client-side execution, and encrypted data transmission.	Role-Based Access Control (itil, incident_manager roles), HTTPS/TLS 1.3, OAuth 2.0, OWASP Top 10 Guidelines
3	Scalable Architecture	Decoupled client-side scripting and multi-tier cloud infrastructure supporting high concurrent user load.	Multi-Tier Cloud Architecture, Client-side browser execution offloading server burden
4	Availability	High-availability cloud hosting with multi-zone redundancy and active-active failover support.	99.99% Cloud Uptime SLA, Multi-Zone Data Center Redundancy, Load Balancers
5	Performance	Sub-second client-side field validation and UI updates without triggering expensive full-page reloads.	Asynchronous GlideAjax requests, client-side caching, optimized JavaScript event handling

References

- C4 Model for Software Architecture: <https://c4model.com/>
- ServiceNow Developer Documentation (Client Scripts & UI Policies): <https://developer.servicenow.com/>
- IBM Cloud Architecture Center: <https://www.ibm.com/cloud/architecture>
- AWS Architecture Center: <https://aws.amazon.com/architecture>
- OWASP Top 10 Web Application Security Risks: <https://owasp.org/www-project-top-ten/>